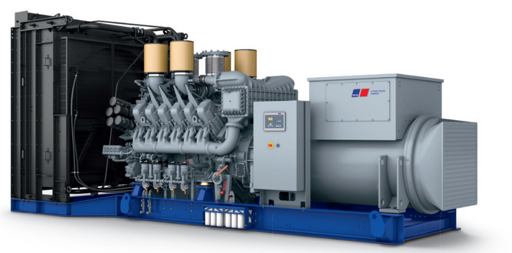
Diesel Generator Set

# mtu 12V4000 DS1750

## 380V – 11 kV/50 Hz/standby power/fuel consumption optimized 12V4000G74F/water charge air cooling


Optional equipment and finishing shown. Standard may vary.

## Product highlights

### Benefits
— Low fuel consumption
— Optimized system integration ability
— High reliability
— High availability of power
— Long maintenance intervals

### Support
— Global product support offered

### Standards
— Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
— Generator set complies to ISO 8528
— Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
— NFPA 110

### Power rating
— System ratings: 1720 kVA - 1880 kVA
— Accepts rated load in one step per NFPA 110*
— Generator set complies to G3 according to ISO 8528-5
— Generator set exceeds load steps according to ISO 8528-5*

### Performance assurance certification (PAC)
— Engine-generator set tested to ISO 8528-5 for transient response
— 85% load factor
— Verified product design, quality and performance integrity
— All engine systems are prototype and factory tested

### Complete range of accessories available
— Control panel
— Power panel
— Circuit breaker/power distribution
— Fuel system
— Fuel connections with shut-off valve mounted to base frame
— Starting/charging system
— Exhaust system
— Mechanical and electrical driven radiators
— Medium and oversized voltage alternators

### Emissions
— Fuel consumption optimized

### Certifications
— CE certification option
— Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

## Application data [1)]

### Engine

| | |
|---|---|
| Manufacturer | *mtu* |
| Model | 12V4000G74F |
| Type | 4-cycle |
| Arrangement | 12V |
| Displacement: l | 57.2 |
| Bore: mm | 170 |
| Stroke: mm | 210 |
| Compression ratio | 16.4 |
| Rated speed: rpm | 1500 |
| Engine governor | ECU 9 |
| Max power: kWm | 1575 |
| Air cleaner | dry |

### Fuel system

| | |
|---|---|
| Maximum fuel lift: m | 5 |
| Total fuel flow: l/min | 16 |

### Fuel consumption [2)]

| | l/hr | g/kwh |
|---|---|---|
| At 100% of power rating: | 358.6 | 189 |
| At 75% of power rating: | 276.1 | 194 |
| At 50% of power rating: | 189.8 | 200 |

### Liquid capacity (lubrication)

| | |
|---|---|
| Total oil system capacity: l | 260 |
| Engine jacket water capacity: l | 160 |
| Intercooler coolant capacity: l | 40 |

### Combustion air requirements

| | |
|---|---|
| Combustion air volume: m³/s | 1.8 |
| Max. air intake restriction: mbar | 50 |

### Cooling/radiator system

| | |
|---|---|
| Coolant flow rate (HT circuit): m3/hr | 56 |
| Coolant flow rate (LT circuit): m3/hr | 30 |
| Heat rejection to coolant: kW | 580 |
| Heat radiated to charge air cooling: kW | 260 |
| Heat radiated to ambient: kW | 75 |
| Fan power for electr. radiator (40°C): kW | 38 |

### Exhaust system

| | |
|---|---|
| Exhaust gas temp. (after turbocharger): °C | 440 |
| Exhaust gas volume: m³/s | 4.5 |
| Maximum allowable back pressure: mbar | 85 |
| Minimum allowable back pressure: mbar | 30 |

## Standard and optional features

### System ratings (kW/kVA)

| Generator model | Voltage | fuel consumption optimized | | | | | |
|---|---|---|---|---|---|---|---|
| | | without radiator | | | with mechanical radiator | | |
| | | kWel | kVA* | AMPS | kWel | kVA* | AMPS |
| Leroy Somer LSA52.3 S5 (Low voltage Leroy Somer standard) | 380 V | 1504 | 1880 | 2856 | 1456 | 1820 | 2765 |
| | 400 V | 1504 | 1880 | 2714 | 1456 | 1820 | 2627 |
| | 415 V | 1504 | 1880 | 2615 | 1456 | 1820 | 2532 |
| Leroy Somer LSA52.3 S6 (Low voltage Leroy Somer oversized) | 380 V | 1504 | 1880 | 2856 | 1456 | 1820 | 2765 |
| | 400 V | 1504 | 1880 | 2714 | 1456 | 1820 | 2627 |
| | 415 V | 1504 | 1880 | 2615 | 1456 | 1820 | 2532 |
| Marathon 743RSL7090 (Low voltage Marathon) | 380 V | 1448 | 1810 | 2750 | 1448 | 1810 | 2750 |
| | 400 V | 1448 | 1810 | 2613 | 1448 | 1810 | 2613 |
| | 415 V | 1376 | 1720 | 2393 | 1376 | 1720 | 2393 |
| Marathon 744RSL7091 (Low voltage Marathon oversized) | 380 V | 1448 | 1810 | 2750 | 1448 | 1810 | 2750 |
| | 400 V | 1448 | 1810 | 2613 | 1448 | 1810 | 2613 |
| | 415 V | 1376 | 1720 | 2393 | 1376 | 1720 | 2393 |
| Marathon 744RSL7091 (Low voltage Marathon engine output optimzed) | 380 V | 1488 | 1860 | 2826 | 1448 | 1810 | 2750 |
| | 400 V | 1496 | 1870 | 2699 | 1448 | 1810 | 2613 |
| | 415 V | 1488 | 1860 | 2588 | 1448 | 1810 | 2518 |
| Marathon 1020FDH7095 (Medium volt. marathon) | 11 kV | 1496 | 1870 | 98 | 1448 | 1810 | 95 |
| Leroy Somer LSA53.2 VL6 (Medium volt. Leroy Somer) | 11 kV | 1496 | 1870 | 98 | 1456 | 1820 | 96 |

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.